

Deep Research: Avatrada 57 Topic 023

Engineering Report: Correlation-Adjusted Position Sizing

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Executive Summary

This report provides a detailed engineering analysis of the "Correlation-Adjusted Sizing" mechanism specified in the source document. The core objective of this mechanism is to manage portfolio risk by reducing the size of a new position based on its correlation with existing assets. A higher correlation implies a smaller diversification benefit, thus warranting a smaller position to control the marginal increase in portfolio volatility.

The analysis deconstructs two approaches: 1. **The Heuristic Model:** A simple, linear reduction factor of $(1 - \text{correlation_coefficient})$, as specified in the source. This method is intuitive but lacks a rigorous mathematical foundation in portfolio theory. 2. **The Rigorous Model:** An approach derived from the foundational Portfolio Variance formula, aiming to maintain a constant level of portfolio risk, as requested by the deep-research prompt.

This report will demonstrate that a direct mathematical solution to "maintain constant portfolio volatility" when *adding* a positively correlated asset is non-trivial and often implies shorting the new asset. A more practical interpretation is to manage the *marginal contribution to risk*.

We will provide a Python implementation for the heuristic model, as it is the most direct interpretation of the source, and critically analyze its limitations, edge cases, and potential improvements for a production environment.

1. Technical Deconstruction

The fundamental principle is that the risk (variance or volatility) of a portfolio is not merely the sum of individual asset risks. The covariance between assets is a critical component. Adding a new asset that is highly correlated with the existing portfolio adds more risk than adding an asset with low or negative correlation.

1.1. Foundational Formula: Portfolio Variance

The analysis begins with the standard formula for the variance of a two-asset portfolio, consisting of existing Asset A and new Asset B.

```
# Math/Pseudocode for Two-Asset Portfolio Variance

Let:
  w_A, w_B = weights (sizes) of Asset A and Asset B
  σ_A, σ_B = volatility (standard deviation of returns) of Asset A and Asset B
  ρ_AB      = correlation coefficient between the returns of A and B

Var(P) = (w_A^2 * σ_A^2) + (w_B^2 * σ_B^2) + (2 * w_A * w_B * ρ_AB * σ_A * σ_B)
```

The final term, $(2 * w_A * w_B * \rho_{AB} * \sigma_A * \sigma_B)$, is the **covariance term**. It shows how the assets' movements interact to either increase ($\rho_{AB} > 0$) or decrease ($\rho_{AB} < 0$) the total portfolio variance.

1.2. The Heuristic Sizing Model (Source Specification)

The source document specifies a simple, linear adjustment factor.

- **Formula:** $\text{adjusted_size} = \text{proposed_size} * (1 - \rho)$
- **Mechanism:** This is a direct and intuitive rule.
 - If $\rho = 1.0$ (perfectly correlated), $\text{adjusted_size} = 0$. The model suggests the new asset adds no diversification and should not be added.
 - If $\rho = 0.6$, $\text{adjusted_size} = \text{proposed_size} * 0.4$. The size is reduced by 60%.

- If $\rho = 0.0$ (uncorrelated), $\text{adjusted_size} = \text{proposed_size} * 1.0$. No adjustment is made.
- If $\rho = -0.5$ (negatively correlated), $\text{adjusted_size} = \text{proposed_size} * 1.5$. The size is increased, rewarding the diversification benefit.

This model is a heuristic, not a direct derivation from the portfolio variance formula. It approximates the desired effect but does not guarantee a specific volatility outcome.

1.3. The Rigorous Sizing Model (Deep Research Prompt)

The prompt asks for a formula to "maintain constant portfolio volatility." Let's analyze this strictly.

- **Initial State:** Portfolio contains only Asset A.
 - $\text{Var}(P_{\text{initial}}) = w_A^2 * \sigma_A^2$
- **Final State:** We add Asset B with an adjusted weight w_{B_adj} .
 - $\text{Var}(P_{\text{final}}) = w_A^2 * \sigma_A^2 + w_{B_adj}^2 * \sigma_B^2 + 2 * w_A * w_{B_adj} * \rho_{AB} * \sigma_A * \sigma_B$
- **Constraint:** $\text{Var}(P_{\text{final}}) = \text{Var}(P_{\text{initial}})$

Setting the two equations equal: $w_A^2 * \sigma_A^2 = w_A^2 * \sigma_A^2 + w_{B_adj}^2 * \sigma_B^2 + 2 * w_A * w_{B_adj} * \rho_{AB} * \sigma_A * \sigma_B$

Simplifying by subtracting $w_A^2 * \sigma_A^2$ from both sides: $0 = w_{B_adj}^2 * \sigma_B^2 + 2 * w_A * w_{B_adj} * \rho_{AB} * \sigma_A * \sigma_B$

Factoring out w_{B_adj} : $w_{B_adj} * (w_{B_adj} * \sigma_B^2 + 2 * w_A * \rho_{AB} * \sigma_A * \sigma_B) = 0$

This gives two possible solutions for w_{B_adj} : 1. $w_{B_adj} = 0$ (The trivial solution: don't add the asset). 2. $w_{B_adj} * \sigma_B^2 + 2 * w_A * \rho_{AB} * \sigma_A * \sigma_B = 0$ $w_{B_adj} = -2 * \rho_{AB} * (w_A * \sigma_A / \sigma_B)$

Analysis of the Rigorous Result: This formula is mathematically correct but operationally problematic for a long-only strategy. If $\rho_{AB} > 0$, the required w_{B_adj} is **negative**. This means to keep portfolio volatility *exactly constant*,

one must **short** the new correlated asset, effectively creating a hedge. This contradicts the goal of *adding* a new position to the portfolio.

Therefore, the prompt's constraint is too strict for practical position sizing. The heuristic $(1 - \rho)$ is a pragmatic simplification that reduces, rather than perfectly neutralizes, the marginal risk contribution.

2. Implementation Strategy

Given the analysis, the most direct and practical implementation is the heuristic model. It captures the spirit of the requirement without the complexities and counter-intuitive results of the strict mathematical derivation.

2.1. Core Function

The implementation is a single Python function. It should include robust input validation.

```
import numpy as np

def adjust_size_for_correlation(
    proposed_size: float,
    correlation: float
) -> float:
    """
    Adjusts a proposed position size based on its correlation to an existing
    portfolio.

    This function implements the heuristic model:  $size * (1 - correlation)$ .
    It reduces size for positive correlation and increases it for negative
    correlation (diversification benefit).

    Args:
        proposed_size (float): The initial desired size of the new position,
            e.g., from a Kelly Criterion calculation.
        correlation (float): The correlation coefficient between the new asset's
            returns and the existing portfolio's returns.
            Must be in the range [-1.0, 1.0].
    """
```

Returns:

float: The adjusted position size. Returns 0.0 if the proposed size is non-positive.

Raises:

ValueError: If correlation is outside the valid [-1.0, 1.0] range.

"""

```
if not -1.0 <= correlation <= 1.0:
```

```
    raise ValueError("Correlation must be between -1.0 and 1.0.")
```

```
if proposed_size <= 0:
```

```
    return 0.0
```

```
# The core adjustment factor from the source specification
```

```
adjustment_factor = 1.0 - correlation
```

```
adjusted_size = proposed_size * adjustment_factor
```

```
# Ensure the adjusted size is not negative (can happen if rho > 1, though validated)
```

```
return max(0.0, adjusted_size)
```

2.2. Broader System Integration

This function is a component within a larger portfolio management or execution system.

1. Input Generation:

- `proposed_size`: This would typically be the output of a primary alpha or sizing model (e.g., Kelly Criterion, fixed fractional sizing).
- `correlation`: This is the most critical input. It should not be the correlation to a single other asset, but to the **entire existing portfolio**.

■ Calculation:

1. Obtain historical return series for the new asset (`R_new`) and the existing portfolio (`R_portfolio`).

2. The portfolio's return series is the weighted average of its components' returns: `R_portfolio(t) = sum(w_i * R_i(t))`.
3. Use a library like `NumPy` or `pandas` to compute the correlation coefficient:
`np.corrcoef(R_new, R_portfolio)[0, 1]`.

2. Workflow:

- An event triggers a potential new trade (e.g., new signal).
- The primary model calculates `proposed_size`.
- The risk management module calculates the `correlation` of the new asset to the current portfolio.
- The `adjust_size_for_correlation` function is called.
- The final `adjusted_size` is sent to the order management system.

3. Critical Analysis

While simple and intuitive, the heuristic model has significant limitations and potential failure modes.

3.1. Potential Failure Modes & Edge Cases

- **Correlation is Not Stable:** The primary failure mode is **regime shift**. Historical correlation, calculated over a specific lookback period, is a poor predictor of future correlation, especially during market stress. In a crisis, correlations across many assets tend to converge towards 1, nullifying diversification benefits precisely when they are most needed. A system relying on this adjustment could take on unintended high risk.
- **Negative Correlation Leverage:** If an asset has a strong negative correlation (e.g., `ρ = -0.8`), the formula increases the position size significantly (`adjusted_size = proposed_size * 1.8`). This introduces leverage. If the correlation regime breaks down and becomes positive, this oversized position becomes a major source of risk. Capping the maximum adjustment factor (e.g., at 1.5x) is a prudent safeguard.

- **Non-Linear Risk Profile:** The linear adjustment $(1 - \rho)$ is a crude approximation of a non-linear (quadratic) problem. It may under-correct risk when adding a very large, moderately correlated position, or over-correct risk for a small position.
- **Multi-Asset Problem:** The prompt simplifies the problem to one existing asset (or portfolio) and one new asset. When considering adding multiple new assets simultaneously, their inter-correlations also matter. Applying this adjustment sequentially can lead to suboptimal portfolio construction.

3.2. Optimizations and Advanced Alternatives

- **EWMA for Correlation:** Instead of a simple moving average correlation, use an Exponentially Weighted Moving Average (EWMA). This gives more weight to recent data, making the correlation estimate more responsive to changing market conditions.
- **Correlation Stress Testing:** Before deploying the `adjusted_size`, the system should run a simulation assuming a "worst-case" correlation scenario (e.g., all ρ values move towards 0.8) to ensure the total portfolio risk remains within acceptable limits.
- **Marginal Contribution to Risk (MCTR):** A more sophisticated approach is to calculate the new asset's MCTR. This measures the precise amount the asset will add to the total portfolio volatility. Sizing can then be adjusted to ensure the new position's risk contribution does not exceed a predefined budget.
- **Full Portfolio Optimization:** The gold standard is to use a mean-variance optimizer. When a new asset is considered, it is added to the existing portfolio, and the optimizer calculates the optimal weights for *all* assets (including the new one) to maximize risk-adjusted return (e.g., Sharpe Ratio) or minimize variance for a given level of expected return. Libraries like `PyPortfolioOpt` or `cvxpy` can implement this. This holistic approach is superior to a pairwise, heuristic adjustment.